Lecture 4

Corey Powell

2026-09-14

Lecture 4

1. Measures of Location (more mathematical notation than the text)
2. Measures of Spread (more mathematical notation than text)
3. Vectors in R
4. Summary Statistics in R
5. Histograms in R
6. Boxplots in R
7. Reading Summary – Chapter 5

Finish reading Chapter 5 and start reading Chapters 8 and 9 for the next lecture.

Measures of Location

- Suppose we have a list of n numbers:

$$x_1, x_2, x_3, \ldots, x_n.$$

- What are some ways we can "summarize" this list with just a single number?
  - Mean
  - Median
  - Mode

Mean

- The mean is just the ordinary average.
- We represent the mean with a bar over the variable.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Mean Example:

Consider the list:

$$3, 8, 4, 9, 1, 1, 2.$$

$$\bar{x} = \frac{1}{7}(3 + 8 + 4 + 9 + 1 + 1 + 2) = 4$$

Median

- The median is the “middle” number in the list.
- First write the list in order, then take the middle number.

Median Example:

Consider the list:

$$3, 8, 4, 9, 1, 1, 2.$$

First put the list in order:

$$1, 1, 2, 3, 4, 8, 9.$$

The median is the middle number: 3.

Median – Technical Definition:

- Minor technical point: What to do if there are an even number of entries in the list?
- One possibility: Use the average of the two middle numbers.
- However, we will not use this convention.
- Below, we will give a precise definition of *percentile*.
- We will define the median as the "50-th percentile".
- In practice, this means the median is the smaller of the middle two numbers.

Percentile – General Idea

- Basic idea: The X-th percentile is the number q such that X percent of the data is less than or equal to q.
- Example: Say a class takes a midterm. The 90th percentile of the midterm scores would be the score where 90% of the class scored lower, and 10% percent higher.
- The exact definition of a percentile is tricky.
  - What if there are only 7 people in the class? There is no test score so that exactly 90% percent get lower, and 10% percent get higher.
  - Another example: What is the 90th percentile of

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10?$$

  - We could choose any number between 9 and 10.

Percentile – Exact Definition

- The Xth percentile of a list is the smallest number q in the list such that at least X percent of the numbers are less than or equal to q.
- Example: Consider the list

$$3, 8, 4, 9, 1, 1, 2.$$

What is the 75th percentile?

Percentile – Exact Definition 2

1. Put the numbers in order:

$$1, 1, 2, 3, 4, 8, 9.$$

2. Each number is $1/7 = 14.23$ percent of the dataset.
   - 28.57% of the numbers are less than or equal to 1.
   - 42.86% of the numbers are less than or equal to 2.
   - 57.14% are less than or equal to 3.
   - 71.43% are less than or equal to 4.
   - 85.71% are less than or equal to 8.
   - 100% are less than or equal to 9.

- The 75th percentile is 8.

Discussion Question – Percentiles

Consider the list:

$$3, 1, 4, 1, 5.$$

1. What is the 100th percentile?
2. What is the 99th percentile?
3. What is the 81st percentile?
4. What is the 80th percentile?
5. What about the 41st, 40th, 39th, 21st, 20th, and 19th?

Remember: The Xth percentile of a list is the smallest number q in the list such that at least X percent of the numbers are less than or equal to q.

Mode

- The most common number in the list.
- There can be more than one. Example: $3, 5, 2, 5, 3$.
- Researchers rarely report the mode.

Mean or Median?

- Consider the list
$$1, 2, 3, 4, 1000.$$
- The mean is 202.
- The median is 3.
- Which number is better?
  - **IT DEPENDS ON CONTEXT!**
- The median is relatively unaffected by extreme values (outliers).
- The real question is, do we *care* about the outliers?

Mean or Median – Examples

- Employee wages at a start-up
- House prices in a city
- Hourly data usage on your cell phone plan

Mathematical Comment 1

Suppose we have a list of n numbers:

$$x_1, x_2, x_3, \ldots, x_n.$$

Consider the quantity

$$\sum_{i=1}^{n} (x_i - a)^2.$$

This quantity is a minimum when a is equal to the mean.

The mean minimizes the total squared deviation.

Mathematical Comment 2

Consider the quantity

$$\sum_{i=1}^{n} |x_i - a|.$$

This quantity is a minimum when a is equal to the median.

The median minimizes the total absolute deviation.

Measures of Spread

- Standard Deviation
- Variance
- Interquartile Range (IQR)
- Median absolute deviation (MAD)

Standard Deviation

$$\mathrm{SD} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Variance

The quantity

$$\mathrm{SD}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

is known as the **variance**.

The variance is theoretically useful, but is not a good measure of spread.

Shortcut Formula for Variance

Note that:

$$\mathrm{Var}(x) = \overline{x^2} - \bar{x}^2$$

The variance is the average of the squares minus the square of the average.

Sometimes this formula can be used as a short-cut to calculate the variance, and thus the standard deviation.

Discussion Question - Shortcut Formula

Consider the list: $1, 2, 3$

Use the shortcut formula to calculate the standard deviation.

Interquartile Range (IQR)

- The IQR is the 75th percentile minus the 25th percentile.
- Despite the name, this is a single number.

Discussion Question - IQR

- What is the IQR of $3, 2, 5, 1, 3$?

Median Absolute Deviation (MAD)

- The median of absolute deviations from the median.

$$\mathrm{MAD} = \mathrm{median}(|x_i - \mathrm{median}(x)|)$$

Discussion Question - MAD

- What is the MAD of $3, 2, 5, 1, 3$?

SD, IQR, or MAD?

- The decision of whether to use the SD, IQR, or MAD is very similar to the decision of whether to use the mean or median.
- The IQR and MAD are both relatively insensitive to outliers.

Vectors in R

- A vector is just a list of numbers.
- In R, you can store the entire vector in one variable.
- We will create vectors in 3 main ways in Stat 280.
- The first is to simply list the numbers manually, using the c command.

```
a = c(1,2,5.7,9.4,0)
print(a)
```

```
## [1] 1.0 2.0 5.7 9.4 0.0
```

Vectors in R (continued)

- The second way is to create a vector filled with a single number repeated many times, using the `rep` command.

```
a = rep(3.2, 6)
print(a)
```

```
## [1] 3.2 3.2 3.2 3.2 3.2 3.2
```

- The final way is to create a sequence of integers using a colon.

```
a = 5:12
print(a)
```

```
## [1]  5  6  7  8  9 10 11 12
```

Indexing

- Once we have a vector of numbers, we can refer to a specific entry in the vector using square brackets.
- For example, to refer to the third entry of the vector a, we'd type a[3]

```
a = 5:12
print(a)
```

```
## [1]  5  6  7  8  9 10 11 12
```

```
print(a[3])
```

```
## [1] 7
```

Summary Statistics in R

- In R, it is very easy to calculate the
  - mean
  - median
  - standard deviation
  - IQR

IQR in R

- The `IQR` function uses a slightly different definition of IQR.
  - The IQR is still the difference between the 75th and 25th percentiles, with the percentiles calculated differently
  - See the link IQR for more information.

```
x = c(3, 8, 4, 9, 1, 1, 2)
```

R Output

```
print(x)
```

```
## [1] 3 8 4 9 1 1 2
```

```
print(mean(x))
```

```
## [1] 4
```

```
n=length(x) #The number of numbers in the vector
#The length, median, SD+, and IQR
print(c(n, median(x),sd(x),IQR(x)))
```

```
## [1] 7.000000 3.000000 3.265986 4.500000
```

```
print(sqrt((n-1)/n)*sd(x)) #This is the SD.
```

```
## [1] 3.023716
```

Review Questions

- See Review_questions3.pdf on Canvas

Histograms in R

Netflix

- `netflix`: Answers to the survey question "How many hours a week do you spend watching Netflix."

```
netflix = survey$q50
print(length(netflix)) #Number of responses
```

```
## [1] 66
```

```
table(netflix) #Frequency table
```

```
## netflix
##    0 0.5    1    2    3    4    5    6    7    9   10   20
##   19    1    4    3    7    8   10    2    6    1    4    1
```

Summary Statistics for Netflix

```
print(mean(netflix))
```

```
## [1] 3.583333
```

```
print(median(netflix))
```

```
## [1] 3
```

```
print(sd(netflix))
```

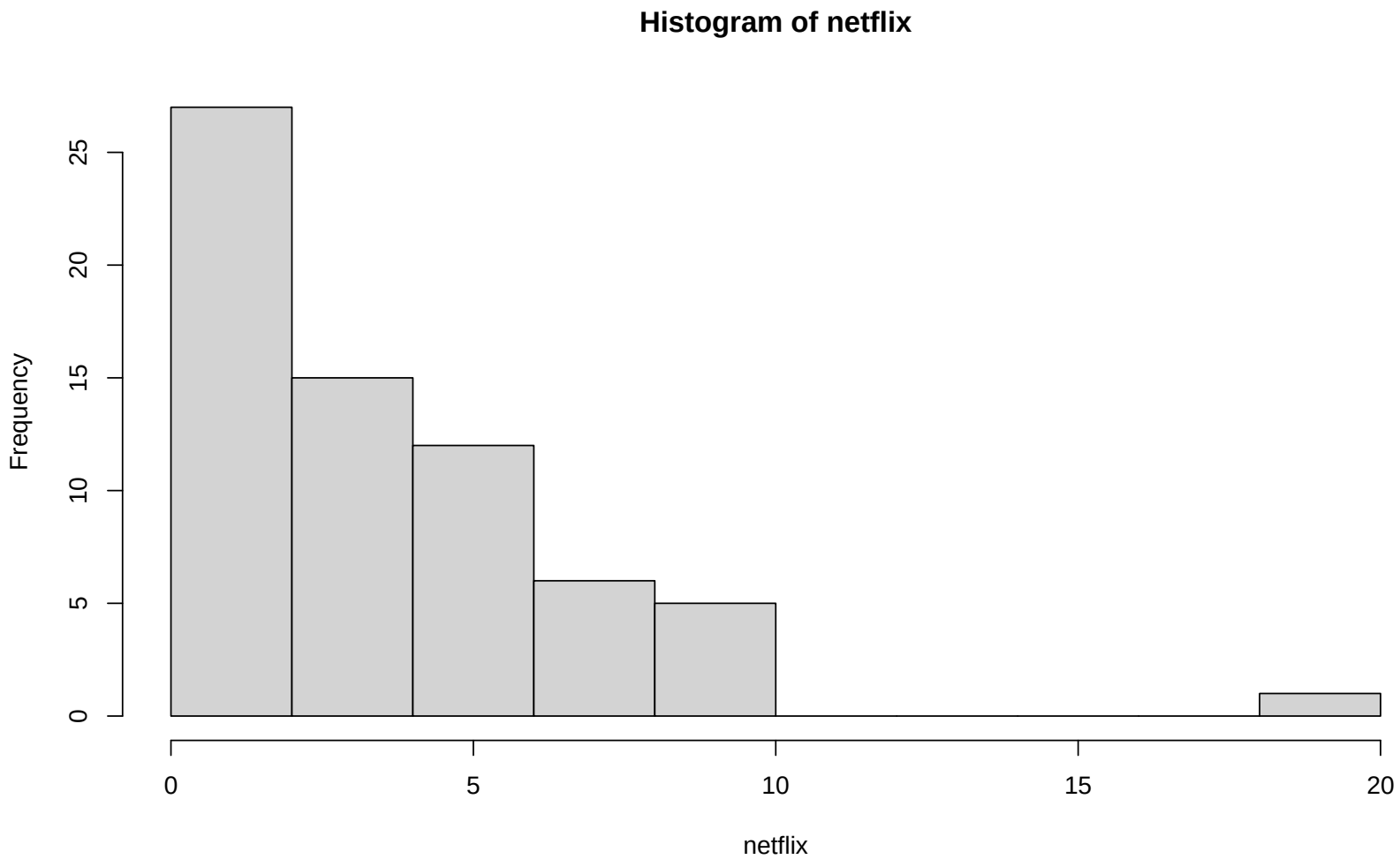
```
## [1] 3.630621
```

```
print(IQR(netflix))
```

```
## [1] 5
```

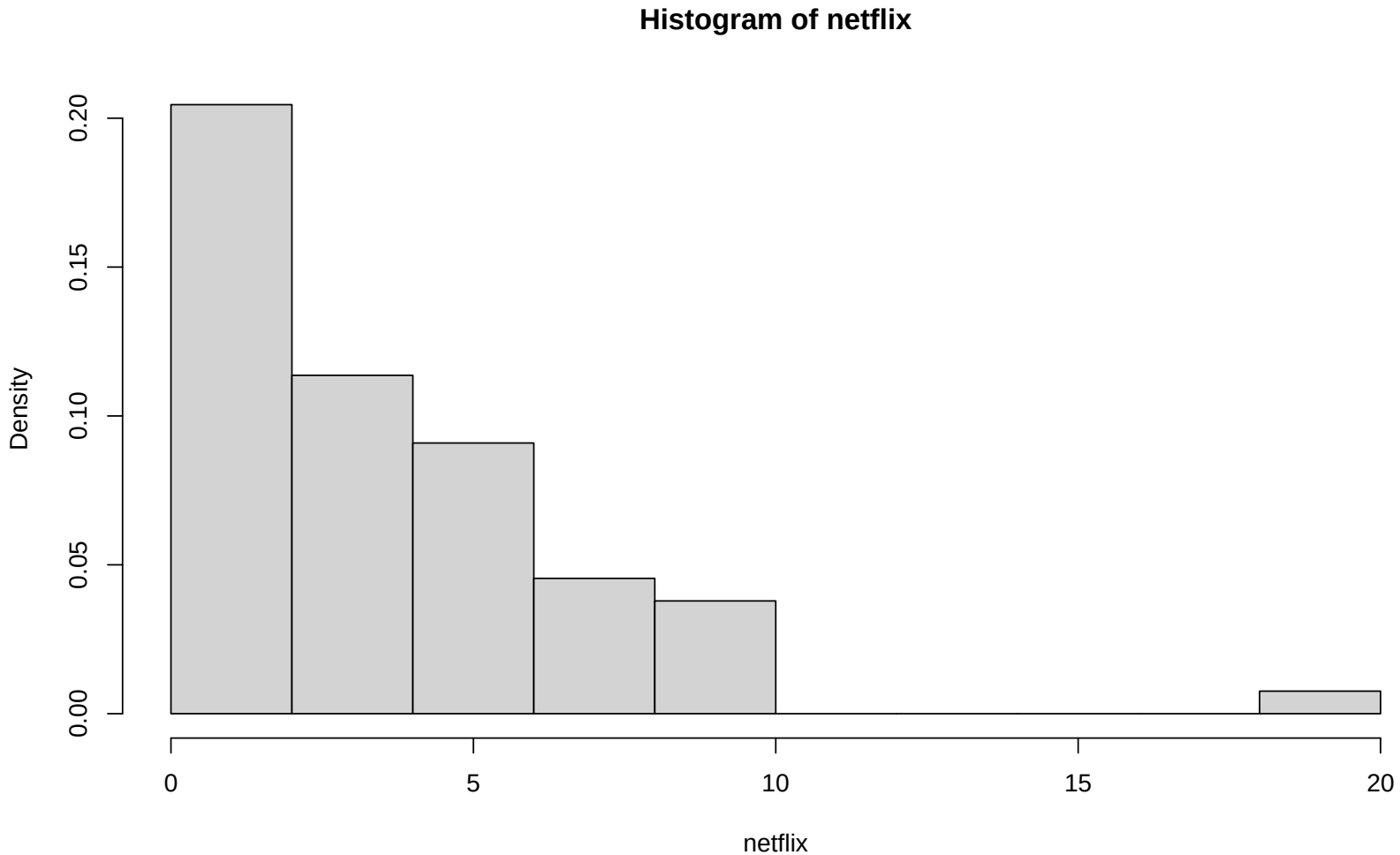
A Frequency Plot for Netflix

```
hist(netflix)  # This is not a histogram
```



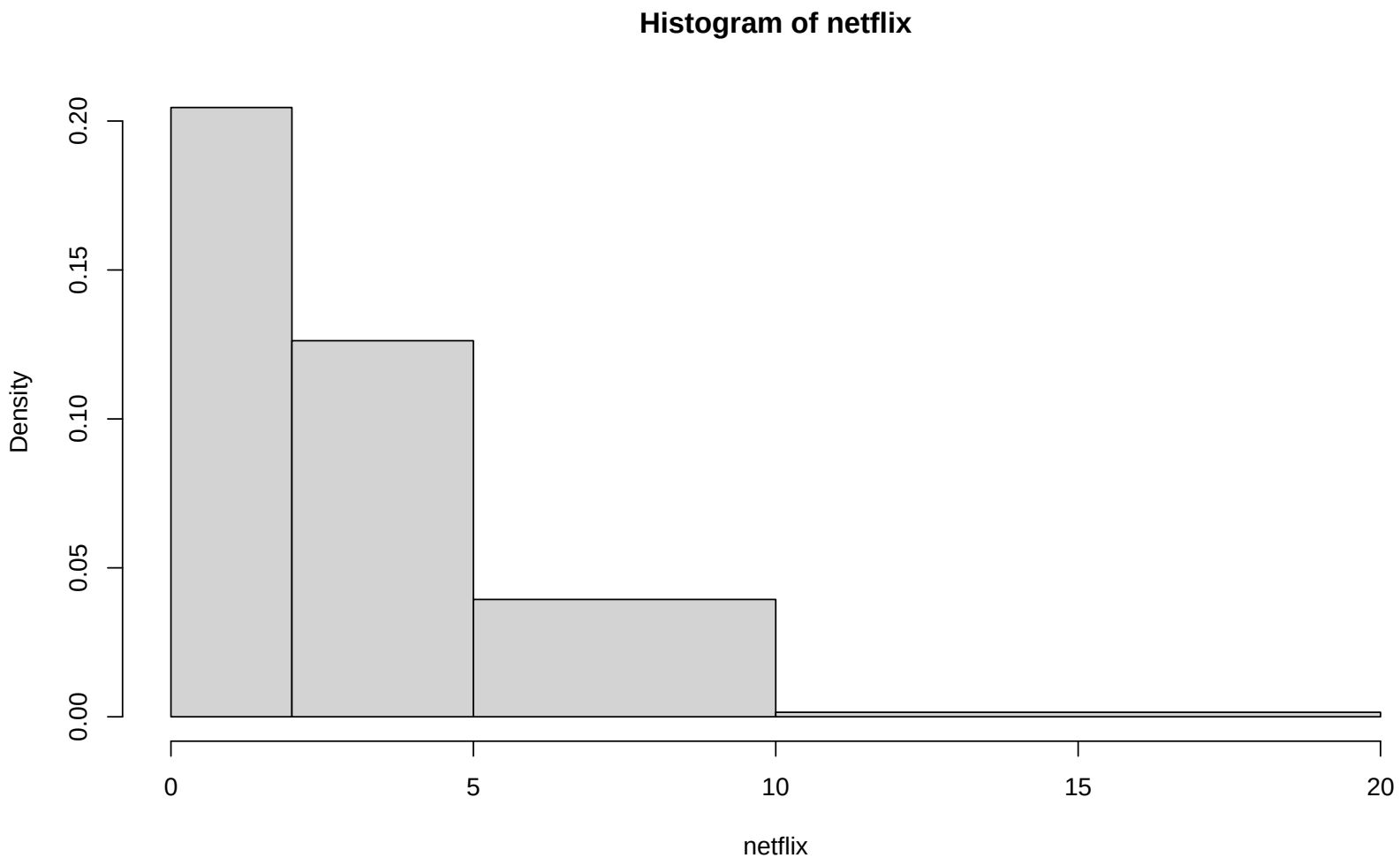
A Histogram for Netflix

```
hist(netflix, freq=FALSE)  # Now it's a histogram
```



Setting the class intervals

```
# Set the class intervals
hist(netflix, freq=FALSE, breaks=c(0, 2, 5, 10, 20))
```



Alternate class intervals

```
mybreaks = 0:21
print(mybreaks[1:11]) #First 11 breaks
```

```
##  [1]  0  1  2  3  4  5  6  7  8  9 10
```

```
print(mybreaks[12:22])#Last 11 breaks
```

```
##  [1] 11 12 13 14 15 16 17 18 19 20 21
```

More Alternate Class Intervals

```
mybreaks = mybreaks - 0.5
print(mybreaks[seq(7)]) #First 7
```

```
## [1] -0.5  0.5  1.5  2.5  3.5  4.5  5.5
```

```
print(mybreaks[seq(8,14)]) #Next 7
```

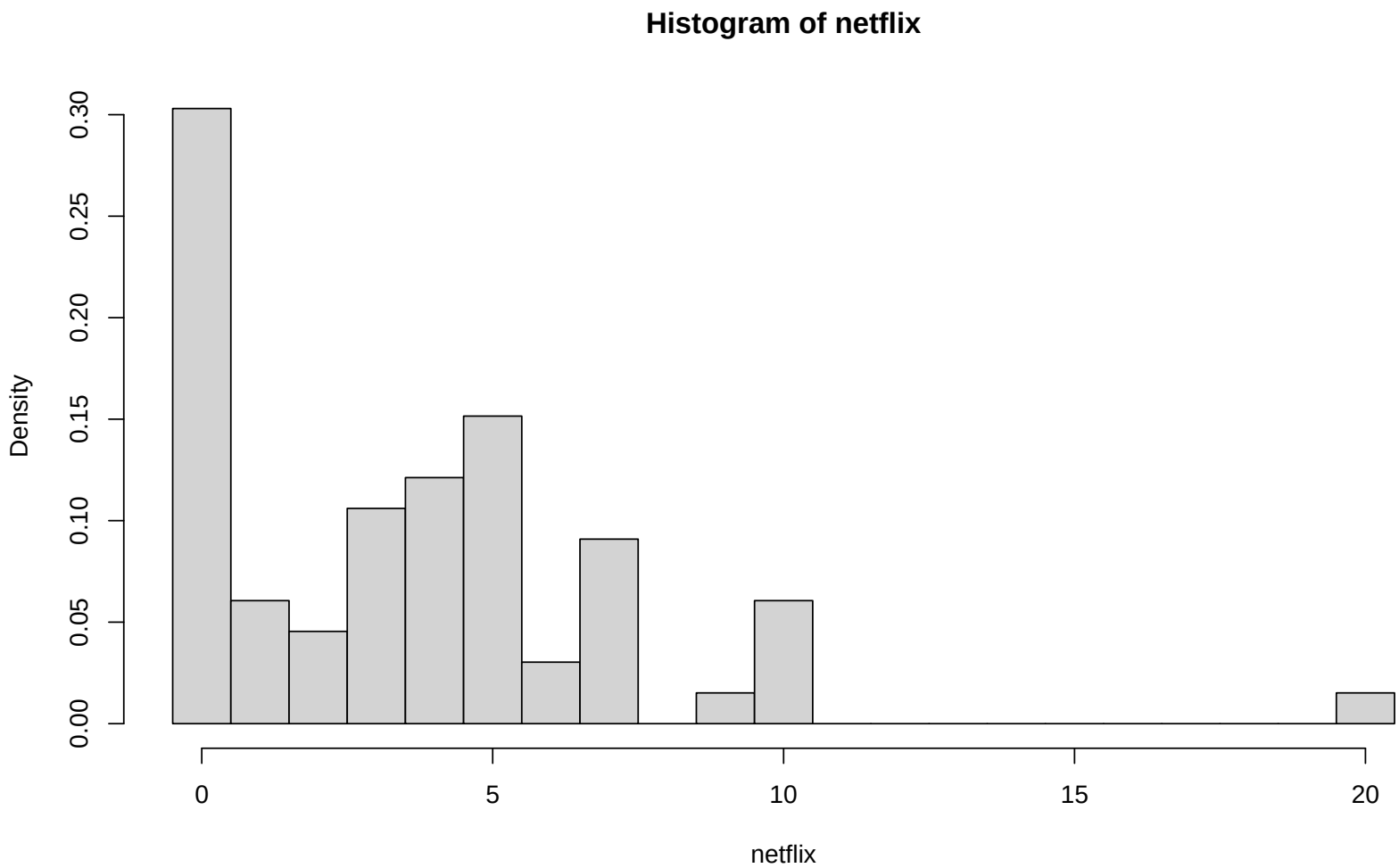
```
## [1]  6.5  7.5  8.5  9.5 10.5 11.5 12.5
```

```
print(mybreaks[seq(15,22)]) #Last 8
```

```
## [1] 13.5 14.5 15.5 16.5 17.5 18.5 19.5 20.5
```

An Alternate Histogram for Netflix

```
# One bar for each integer
hist(netflix, freq=FALSE, breaks=mybreaks)
```



R's Endpoint Convention

- By default, the histogram cells are intervals of the form $(a, b]$.
  - They include their right-hand endpoint, but not their left one, with the exception of the first cell.
- The option `right = FALSE` changes the intervals to the form $[a, b)$.

Making Histograms in R – Summary

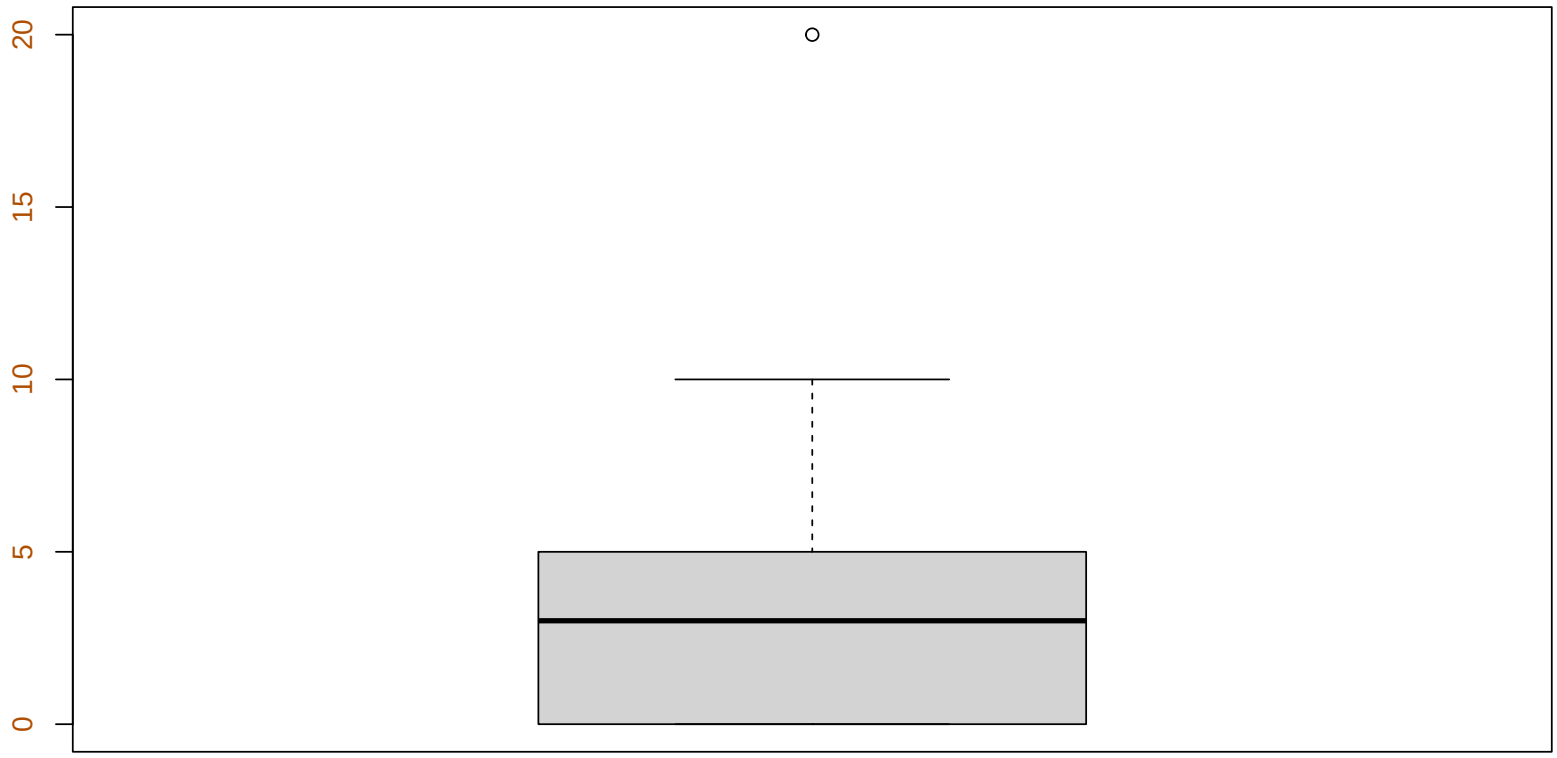
- Use the `hist` command.
- Use the `freq=FALSE` option to get a density scale.
- Often you will want to set the `breaks` yourself, especially if the data is integer-valued.
- Be aware of the endpoint convention.

Boxplots

- Boxplots, like histograms, are a graphical way to summarize a list of numbers.
- Boxplots consist of:
  - A box that shows the 25th, 50th, and 75th percentiles
  - Whiskers that capture the range of most of the rest of the data
  - Individual extreme values (outliers), if present

A Boxplot for Netflix

```
netflix = survey$q50
boxplot(netflix)
```

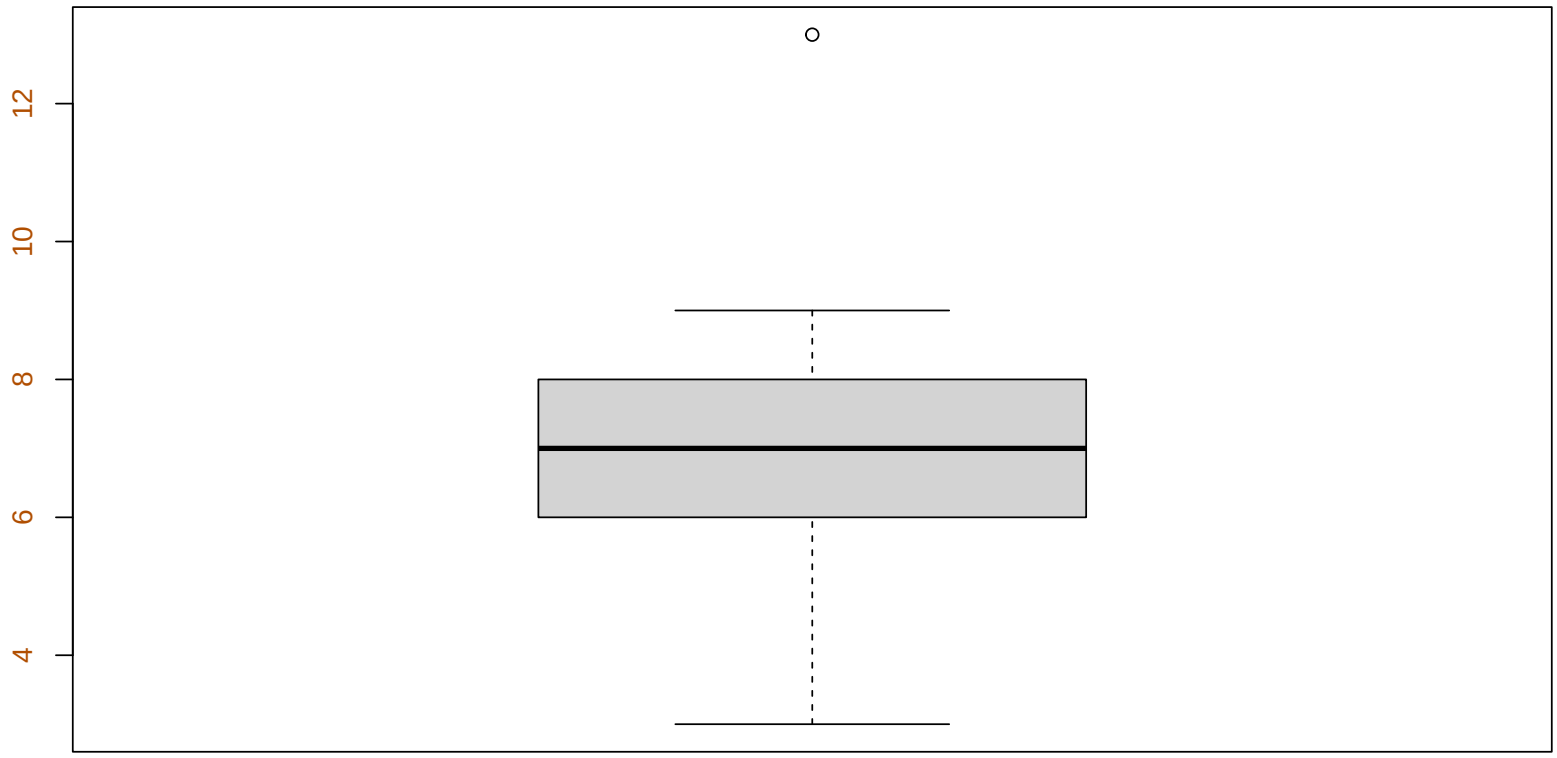


How far out do the whiskers go?

- Unfortunately, there is no real consensus. Some common strategies (from Wikipedia):
  - The minimum and maximum of all of the data
  - The lowest datum still within 1.5 IQR of the lower quartile, and the highest datum still within 1.5 IQR of the upper quartile (often called the Tukey boxplot)
  - One standard deviation above and below the mean of the data
  - The 9th percentile and the 91st percentile
  - The 2nd percentile and the 98th percentile.
- By default, R makes a Tukey boxplot.

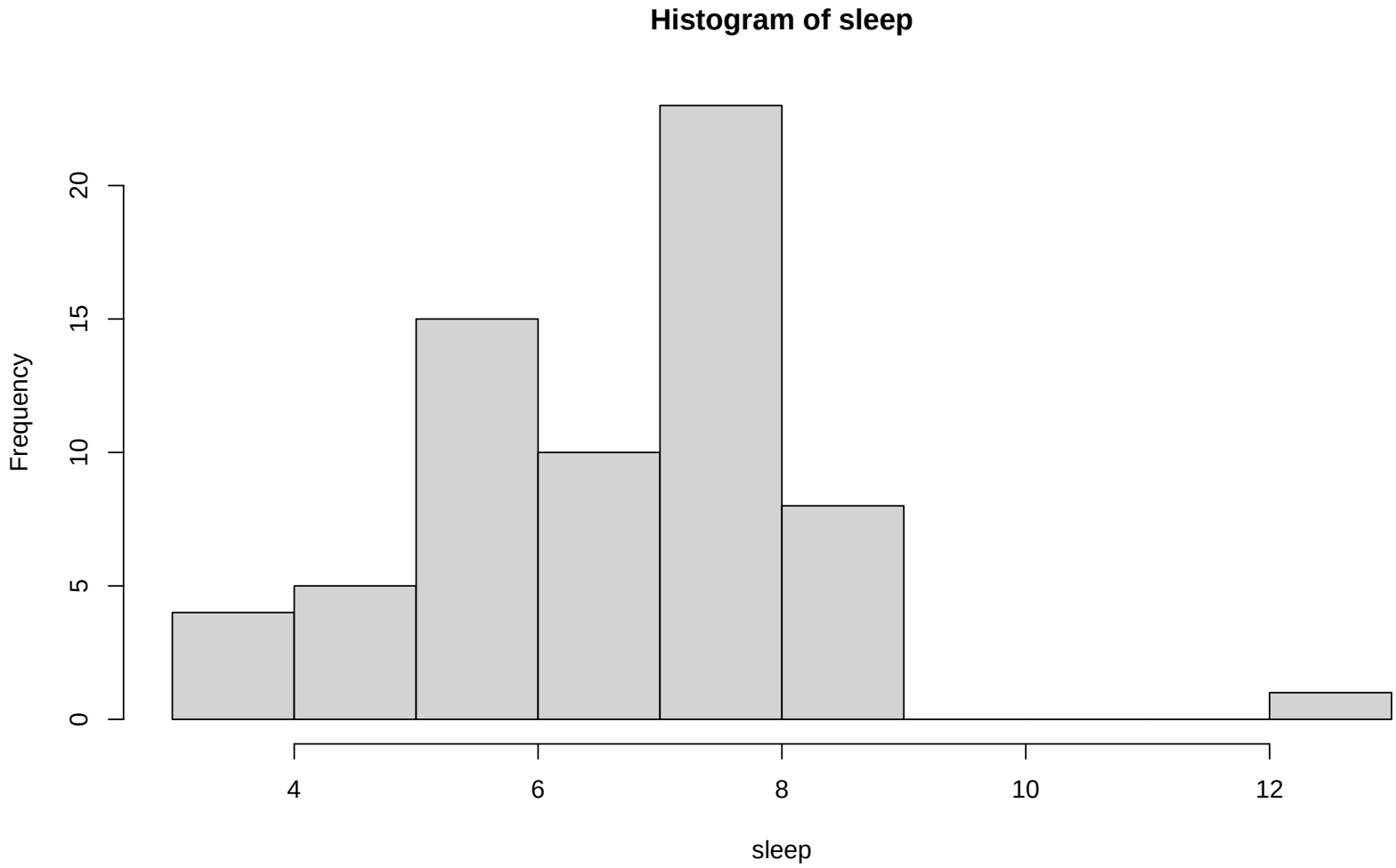
A Boxplot for Hours of Sleep Last Night

```
sleep = survey$q55
boxplot(sleep)
```



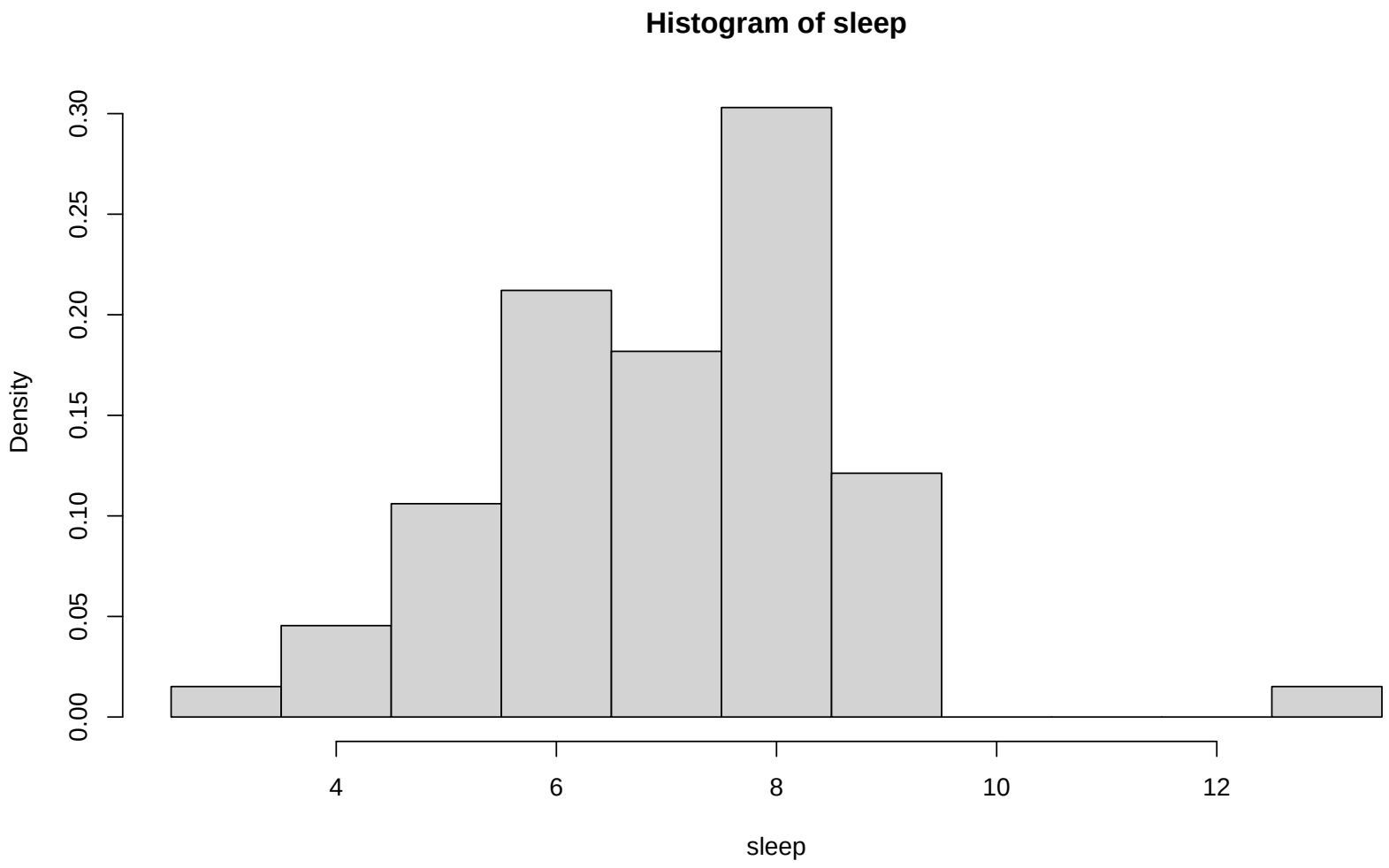
A Frequency Plot for Sleep

```
hist(sleep)
```



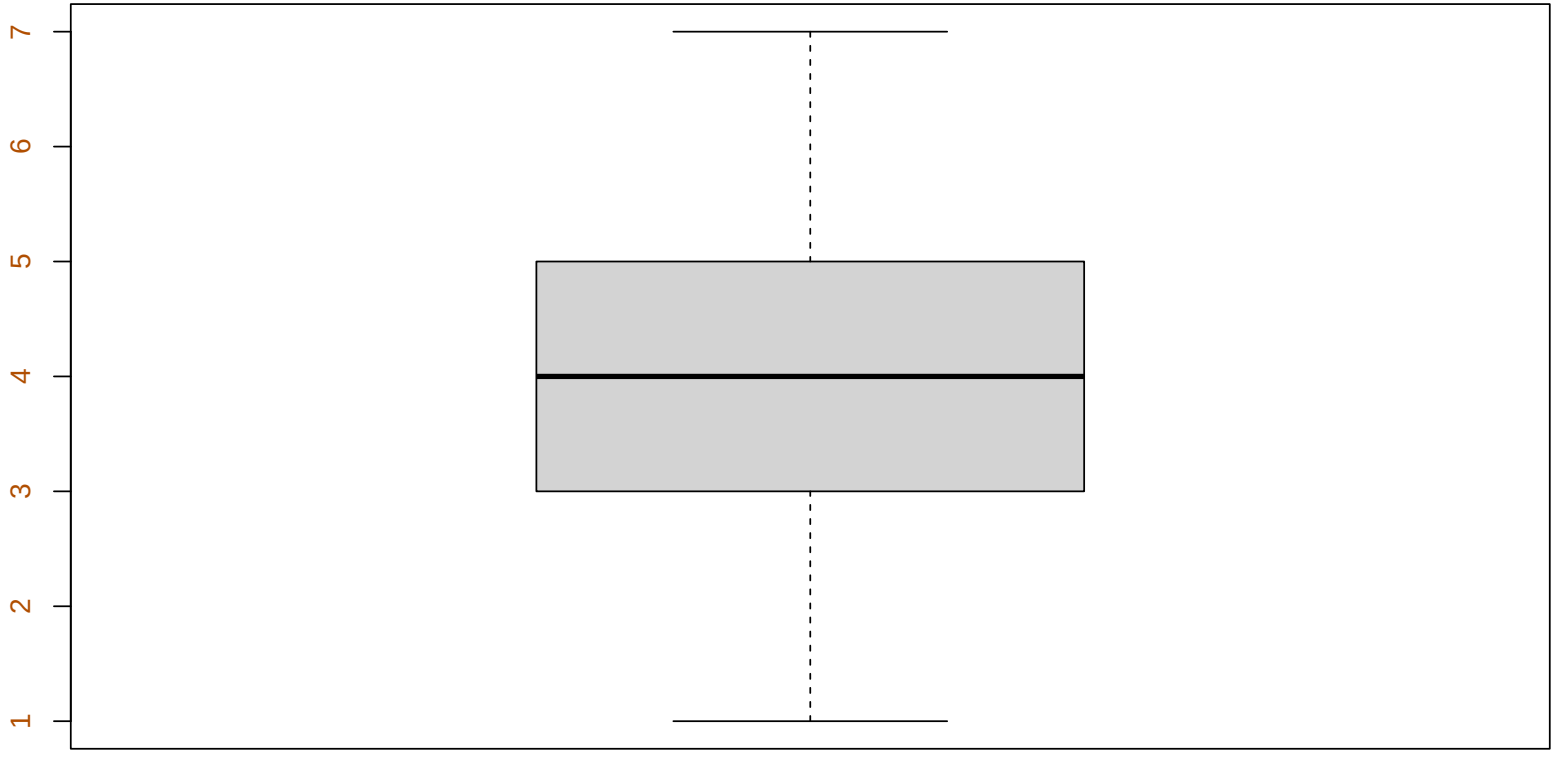
A Histogram for Sleep

```
mybreaks = 2:13 + 0.5
hist(sleep, freq=FALSE, breaks=mybreaks)
```



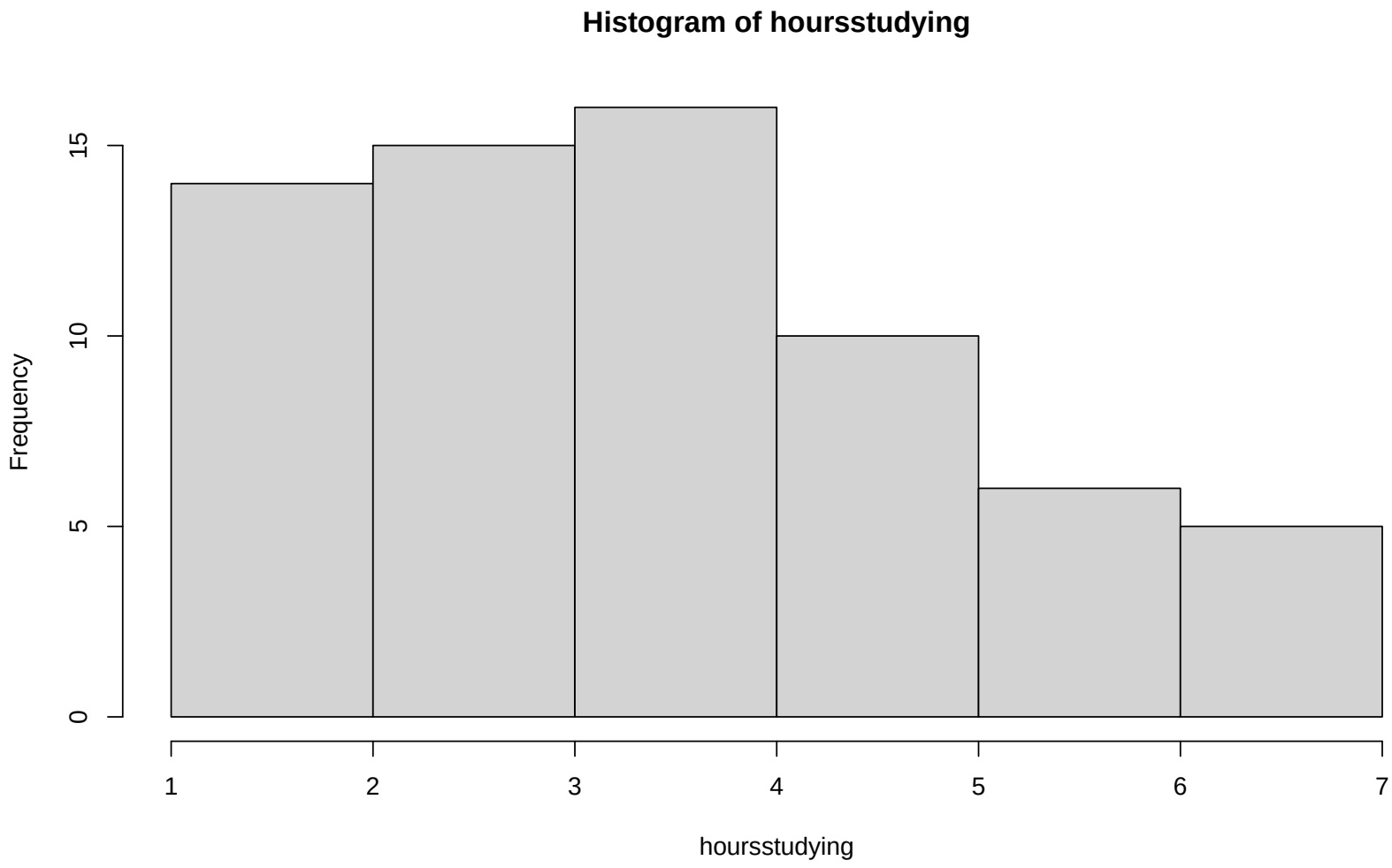
Visualizing Hours Studying Per Night

```
hoursstudying = survey$q100
boxplot(hoursstudying)
```



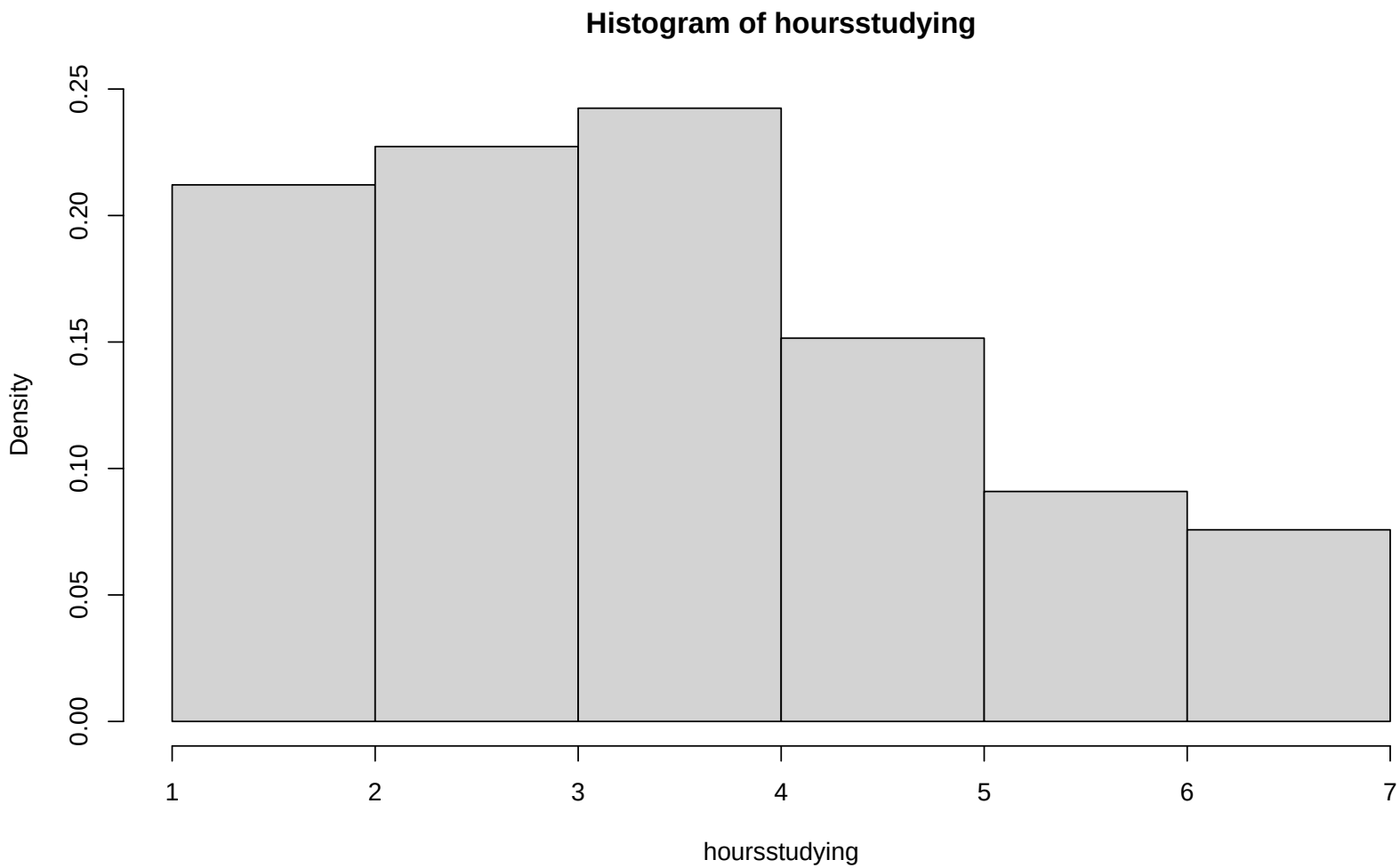
A Frequency Plot for Hours Studying

```
hist(hoursstudying)
```



A Histogram for Hours Studying

```
# What modifications should we make?
hist(hoursstudying, freq=FALSE)
```

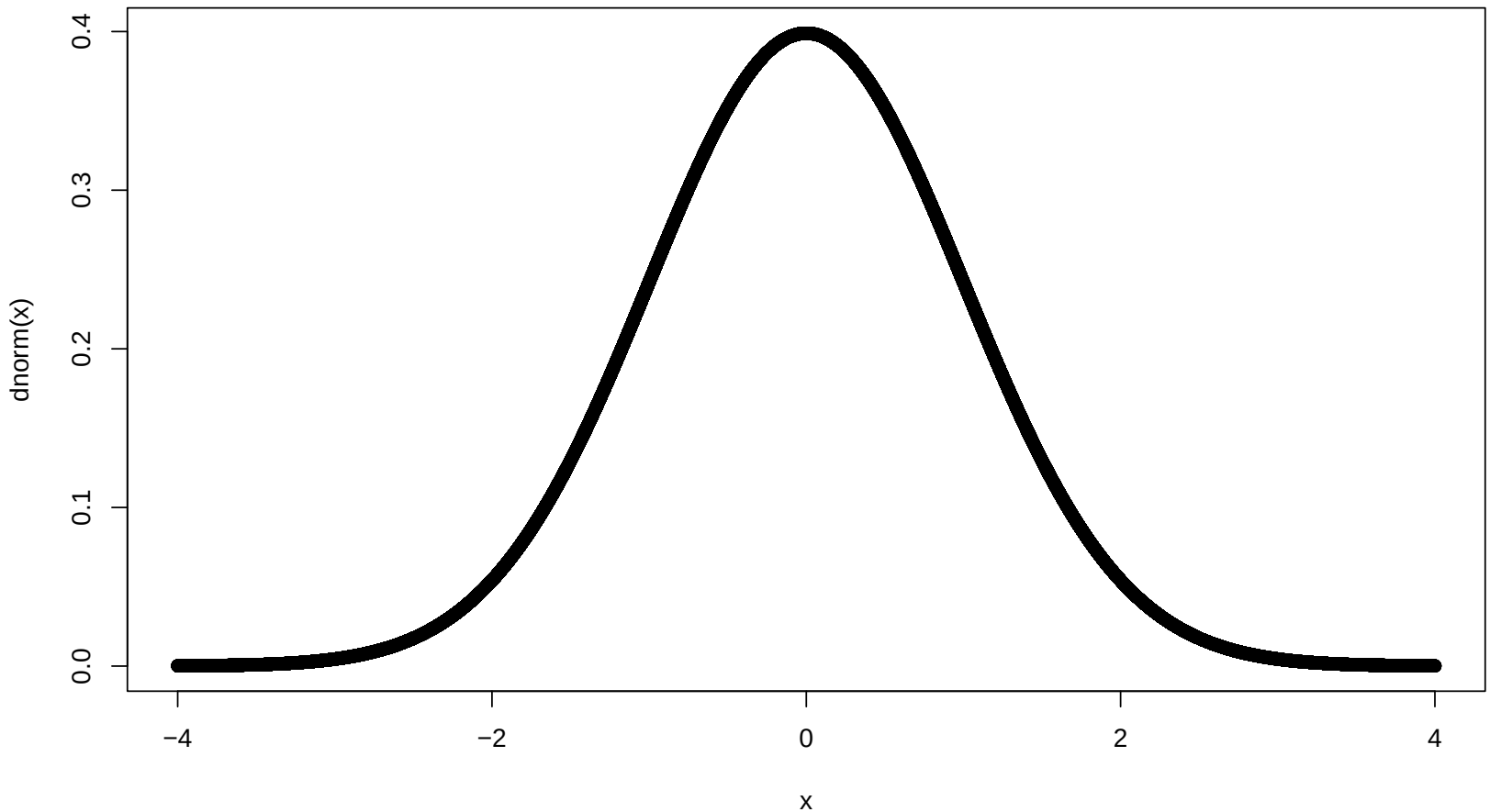


Reading Summary – Chapter 5

Section 5.1: The Normal Curve

- You can convert a value to **standard units** by calculating how many SDs it is above or below average.
- $\text{Value in Standard Units} = \dfrac{\text{Value} - \text{Mean}}{\text{Standard Deviation}}$
- $z_i = \dfrac{x_i - \bar{x}}{\text{SD}(x)}$ $\longleftarrow$ This notation not in the book

The Normal Curve



Section 5.2: Finding Areas Under the Normal Curve

- The Normal Curve is an idealized histogram. The area under it is 100%.
- To find the area under the normal curve in some interval, use the table in the back of the book.
- Outside of homework and exams, you will find areas using the `pnorm` function in `R`.
- The value `pnorm(x)` represents the percentage of the area under the normal curve that is to the left of `x`.

Examples

- Example 1: What is the area under the normal curve from -1 to 2?
  - Looking in the table, we see that the area between -1 and 1 is 68%.
  - We also see that the area between -2 and 2 is 95%.
  - Thus the area between -1 and 0 is 34%, and the area between 0 and 2 is 47.5%,
  - so the area between -1 and 2 is $34\% + 47.5\% = 81.5\%$.
- Example 2: What is the area between 1 and 2?
  - Answer: $47.5\% - 34\% = 13.5\%$

Using R

```
pnorm(2) - pnorm(-1) #Area between -1 and 2
```

```
## [1] 0.8185946
```

```
pnorm(2) - pnorm(1)  #Area between 1 and 2
```

```
## [1] 0.1359051
```

Section 5.3: The Normal Approximation for Data

- It is a remarkable fact that many histograms follow the normal curve.
- If a histogram approximately follows the normal curve, then you can approximate areas in the histogram by areas under the normal curve.
- To approximate the area under a histogram between values x_1 and x_2:
  - Convert x_1 and x_2 to standard units z_1 and z_2.
  - Find the area under the normal curve between z_1 and z_2 using the table.

Section 5.4: Percentiles

- Many histograms do not follow the normal curve.
- Introduction to percentiles
- Knowing the percentiles $\longleftrightarrow$ Knowing the histogram $\longleftrightarrow$ Knowing the list

Section 5.5: Percentiles and the Normal Curve

- If you know the mean and standard deviation, and the histogram follows the normal curve, then you can calculate the percentiles using the normal table.

Section 5.6: Change of Scale

- If you add a constant c to every number in a list, then
  - The average goes up by the constant:
    $\text{new average} = \text{old average} + c$.
  - The SD doesn't change: $\text{new SD} = \text{old SD}$
- If you multiply every number in a list by a constant, then
  - The average gets multiplied by the constant:
    $\text{new average} = (\text{old average}) \times c$
  - The SD gets multiplied by the absolute value of the constant:
    $\text{new SD} = (\text{old SD}) \times |c|$

Review questions

- See Review_questions4.pdf for review questions from this lecture and the first part of the next lecture.